ONLINE EXAM AND MANAGEMENT SYSTEM

1.Case Study

Introduction: The *Online Examination Management System* is a digital platform developed to conduct examinations through the internet in an efficient and organized manner. In traditional examination systems, institutions conduct exams using paper-based methods, which require printing question papers, distributing them to students, manually checking answer sheets, and preparing results. This process consumes a lot of time, effort, and resources. The Online Examination Management System solves these problems by automating the entire examination process and storing all information in a centralized database.

The main objective of this system is to simplify the management of examinations for educational institutions such as schools, colleges, and training centers. The system allows administrators to create and manage exams, add subjects and questions, and monitor student performance. Students can log in to the system, attempt online exams, submit answers, and view their results instantly. By using a database management system, the system can securely store and manage large amounts of student, exam, and result data.

The system mainly consists of two types of users: *Admin* and *Student*. The administrator manages the entire system by creating exams, adding questions, managing subjects, and publishing results. Students register in the system and log in using their credentials. After logging in, students can select available exams, answer questions within a given time limit, and submit their responses. Once the exam is completed, the system automatically evaluates the answers and generates the result.

The database of the Online Examination Management System includes important entities such as *Student, Admin, Subject, Examination, Questions, and Result*. Each entity stores specific information related to the system. For example, the student entity stores student details such as ID, name, contact information, and qualification. The examination entity stores information about exams, while the questions entity stores the questions and correct answers for each exam. The result entity records the marks and grades obtained by students.

This system provides many advantages, such as reducing paperwork, saving time, and providing quick and accurate results. It also improves the efficiency of exam management and ensures secure storage of information. However, the system requires a stable internet connection and proper security measures to prevent unauthorized access.

In conclusion, the Online Examination Management System is a useful and efficient solution for conducting examinations digitally. It helps institutions manage exams easily, improves accuracy in result generation, and provides a convenient platform for students to take exams online.

2.OBJECTIVE OF THE SYSYEM

The main objectives of the Online Examination Management System are:

To automate the examination process.

To reduce paperwork and manual effort.

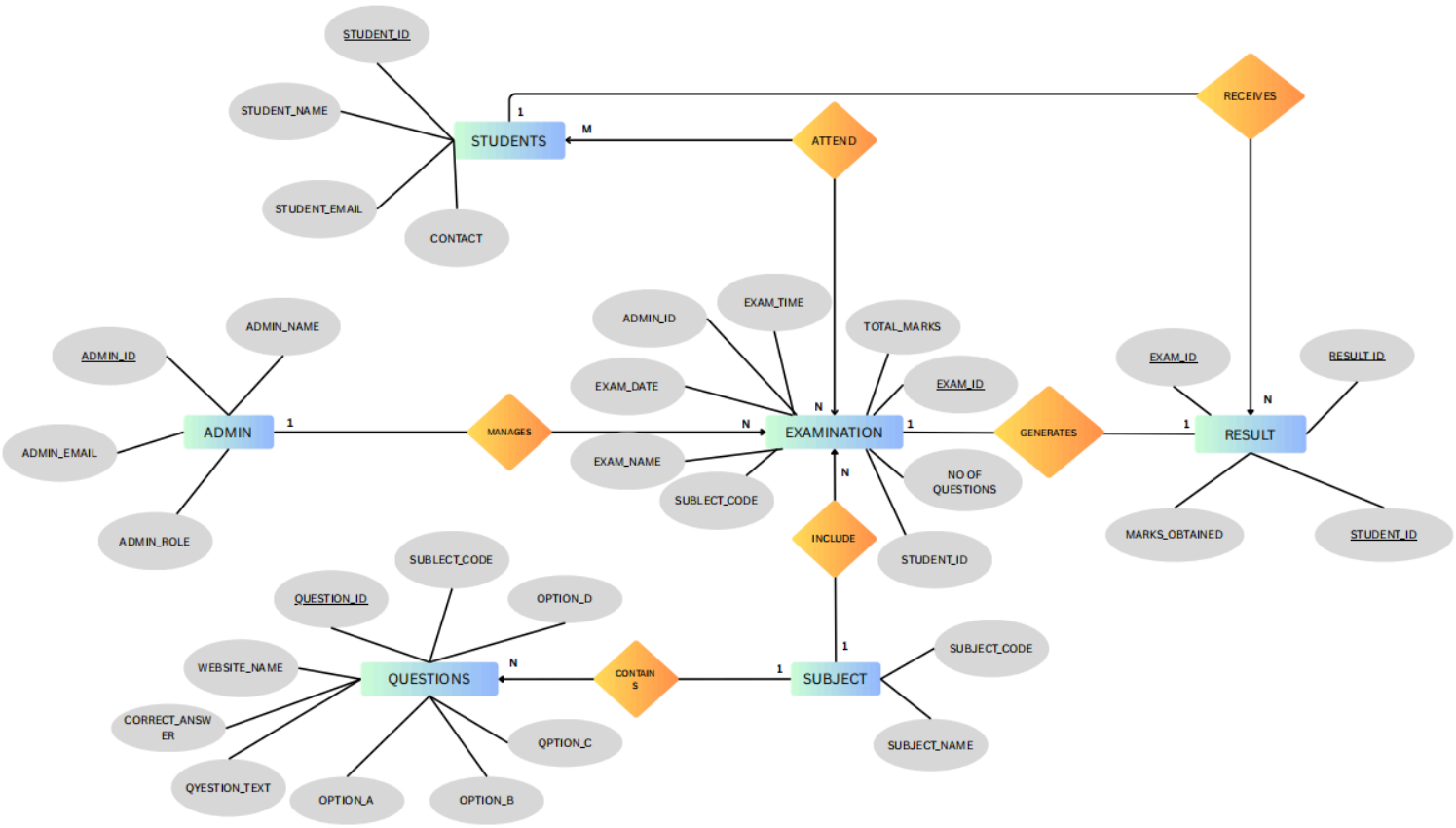
To provide quick and accurate results.

To manage student and exam records efficiently.

To allow students to take exams from anywhere.

To maintain secure and organized exam data in a database.

3.ER DIAGRAM (Entity realtionship)



4.RESOURCE TABLE

STUDENTS		
Field Name	**Data Type**	**Description**
STUDENT_ID	INT (PK)	Unique student ID
STUDENT_NAME	VARCHAR(100)	Student name
STUDENT_EMAIL	VARCHAR(100)	Unique student email
CONTACT	VARCHAR(15)	Contact number

ADMIN		
Field Name	**Data Type**	**Description**
ADMIN_ID	INT (PK)	Unique admin ID
ADMIN_NAME	VARCHAR(100)	Admin name
ADMIN_EMAIL	VARCHAR(100)	Unique admin email
ADMIN_ROLE	VARCHAR(50)	Admin role

SUBJECT		
Field Name	**Data Type**	**Description**
SUBJECT_CODE	VARCHAR(20) (FK)	Unique subject code
SUBJECT_NAME	VARCHAR(100)	Subject name

QUESTIONS		
Field Name	**Data Type**	**Description**
QUESTION_ID	INT (PK)	Unique question ID
SUBJECT_CODE	VARCHAR(20) (FK)	Subject reference
QUESTION_TEXT	TEXT	Question text
OPTION_A	VARCHAR(255)	Option A
OPTION_B	VARCHAR(255)	Option B
OPTION_C	VARCHAR(255)	Option C
OPTION_D	VARCHAR(255)	Option D
CORRECT_ANSWER	VARCHAR(255)	Correct answer
WEBSITE_NAME	VARCHAR(100)	Source website name

EXAMINATION		
Field Name	**Data Type**	**Description**
EXAM_ID	INT (PK)	Unique exam ID
EXAM_NAME	VARCHAR(150)	Exam name
EXAM_DATE	DATE	Exam date
EXAM_TIME	TIME	Exam time
TOTAL_MARKS	INT	Total marks
ADMIN_ID	INT (FK)	Number of questions
SUBJECT_CODE	VARCHAR(20) (FK)	Subject reference
STUDENT_ID	INT (FK)	Student reference

RESULT		
Field Name	**Data Type**	**Description**
RESULT_ID	INT (PK)	Unique result ID
EXAM_ID	INT (FK)	Exam reference
STUDENT_ID	INT (FK)	Student reference
MARKS_OBTANED	INT	Marks obtained

5.SCOPE OF THE SYSTEM

The Online Examination Management System can be used by schools, colleges, and training institutes to conduct online tests and examinations.

The system includes the following functionalities:

Student registration and login

Admin management of exams

Adding and managing questions

Conducting online tests

Automatic result calculation

Result storage and retrieval

The system ensures secure access so that only authorized users can access exam information.

6.ADVANTAGE OF THE SYSTEM

Saves time and reduces manual work

Quick evaluation of answers

Instant result generation

Easy data management using database

Secure storage of exam information

Accessible from anywhere with internet

7.Primary Keys and Foreign Keys

Primary Keys

Student_ID

Admin_ID

Subject_Code

Exam_ID

Question_ID

Result_ID

Foreign Keys

Subject_Code → SUBJECT

Student_ID → STUDENTS

Exam_ID → EXAMINATION

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